NETWORKING 02

🌐 What is OSI Model?

OSI = Open Systems Interconnection

👉 OSI = **7 layers of how data travels from your computer to another**

Think like:

📦 Sending a parcel
Each layer = one step in the delivery process

Layer	Name
7	Application
6	Presentation
5	Session
4	Transport
3	Network
2	Data Link
1	Physical

🍕 OSI Model = Ordering Pizza

👉 You order pizza from Domino's and it reaches your house
That full process = **OSI model**

🧾 7. Application Layer (You ordering pizza)

👉 You open app and order pizza

📌 Simple:

> "I want pizza 🍕"

💡 In networking:
You use WhatsApp / browser

🔐 6. Presentation Layer (Packing & safety)

👉 Pizza is packed properly

- Covered (safe)
- Nicely arranged

📌 Simple:

> "Make it safe & neat"

💡 In network:
Encryption + formatting

📞 5. Session Layer (Call connection)

👉 Shop confirms your order

- Starts order
- Keeps track
- Ends after delivery

📌 Simple:

> "Connection active till pizza arrives"
>
> # 🚚 4. Transport Layer (Delivery process)
>
> 👉 Pizza may come in parts
>
> - Pizza + drink + sides
> - Ensures all items reach
>
> 📌 Simple:
>
> > "Deliver everything correctly"
>
> 💡 TCP = safe delivery
> 💡 UDP = fast but no guarantee

🗺️ 3. Network Layer (Finding your house)

👉 Delivery guy uses map

- Finds best route
- Uses your address

📌 Simple:

> "Where is your house?"

💡 IP address = your home address

🧭2 . Data Link Layer (Your street/local area)

👉 Now he is in your area

- Finds exact building
- Uses house number

📌 Simple:

> "Which exact house?"

💡 MAC address = exact device

⚡ 1. Physical Layer (Actual delivery)

👉 Pizza reaches your hand 🍕

📌 Simple:

> "Finally delivered physically"

💡 In network:
Signals, cables, WiFi

REAL CASE OF MESSAGING NETWORK

🧾 7. Application Layer (WhatsApp itself)

👉 You type and hit send

- WhatsApp app is working
- Message created

📌 Simple:

> "I want to send this message"
>
> ## 🔐 6. Presentation Layer (Encryption magic 🔒)
>
> 👉 WhatsApp encrypts your message
>
> - Converts "Hi bro" → random coded data
> - So no hacker can read
>
> 📌 Simple:
>
> > "Hide the message so only friend can read"

📞 5. Session Layer (Connection setup)

👉 WhatsApp ensures connection

- You are online
- Friend is reachable
- Chat session exists

📌 Simple:

> "Make sure both sides are connected"

🚚 4. Transport Layer (Breaking message into packets)

👉 Message is broken into small parts

- Sent piece by piece
- Ensures delivery

📌 Simple:

> "Send message safely in parts"

💡 WhatsApp uses TCP → reliable delivery

FTP, TCP, UDP are NOT same type

Name	Type	Role
FTP	Application (top layer)	Used to transfer files
TCP	Transport layer	Ensures safe delivery
UDP	Transport layer	Fast but no guarantee

📌 Simple:

> FTP = *What you want to do*
> TCP/UDP = *How data is sent*
>
> # 📦 FTP = "I want to send a FILE"
>
> 👉 Like saying:
>
> > "Send this package to my friend"
>
> - Upload/download files
> - Works like Google Drive / file sharing
>
> 📌 FTP decides:
>
> - What file
> - Where to send
> - Download/upload rules

🚚 TCP = “Safe Delivery Service”

👉 Like:

- Courier guy calls and confirms
- Ensures package reaches
- Resends if lost

📌 Simple:

> “Deliver properly, no loss”

✔️ Used by FTP

⚡ UDP = “Fast but careless delivery”

👉 Like:

- Throw package and go 😅
- No checking

📌 Simple:

> “Fast but no guarantee”

✔️ Used in:

- Video calls
- Gaming

🧠 Easy Memory Trick

👉 Think:

- FTP = Manager 📂
- TCP = Responsible worker ✅
- UDP = Speedy but careless worker ⚡

🗺️ 3. Network Layer (Finding your friend)

👉 Internet finds your friend's phone

- Uses IP address
- Chooses best route

📌 Simple:

> "Where is your friend on internet?"
>
> ## 2.Data Link Layer (Local network step)
>
> 👉 Data reaches correct device in network
>
> - Uses WiFi / MAC address
>
> 📌 Simple:
>
> > "Send to exact phone in that network"

1.Physical Layer (Actual signal travel)

👉 Real transmission happens

- Signals travel via:
  - WiFi
  - Mobile data
  - Cables

📌 Simple:

> "Actual movement of data"
>
> OSI = "How WhatsApp message secretly travels from your phone to your friend's phone step by step"

🔐 What is SSL? (Super Simple)

SSL = Secure Sockets Layer

🧠 Easy example (real life)

Imagine:

- Without SSL → like sending a **postcard** 📬 (anyone can read it)
- With SSL → like sending a **locked envelope** ✉️🔒 (only receiver can read it)

🌐 Where do you see SSL?

👉 When you open a website like:

- `https://google.com`

✔️ The **"S" in HTTPS = SSL (actually TLS, updated version)**
✔️ That 🔒 lock symbol in browser = SSL protection

🔄 What does SSL actually do?

1️⃣ Encryption 🔐

It converts your data into secret code
👉 So hackers **cannot read it**

Example:
Your password → becomes unreadable text

2️⃣ Authentication ✔️

Checks that website is real
👉 Not a fake website

3️⃣ Data Integrity 📦

Ensures data is not changed while traveling

📶 SSL in OSI Model

👉 SSL works at:

👉 **Layer 6 (Presentation Layer)**
Because it **encrypts & decrypts data**

OSI Model	TCP/IP Model
7 layers	4 layers
Theoretical	Practical
Used for learning	Used in real networks
Very detailed	More simplified